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Welcome to Applied Robotics (FRTN85/FRTN80) !

LECTURE 1

BJÖRN OLOFSSON, DEPT. AUTOMATIC CONTROL, LTH



Today's Lecture

- Welcome to the courses in Applied Robotics!
- Course administration
- Outline of the course
- Introduction to robotics and application areas



Course Administration



Applied Robotics – FRTN85

Applied Robotics for Architects – FRTN80

Course coordinator and examiner

Björn Olofsson

Dept. Automatic Control, M-building (2nd floor)

bjorn.olofsson@control.lth.se



Course administrator

Mika Nishimura

Dept. Automatic Control, M-building (3rd floor)

mika.nishimura@control.lth.se

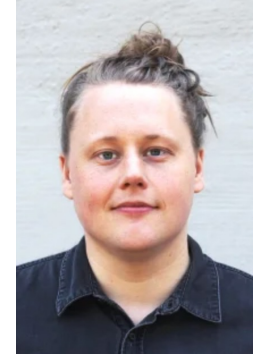
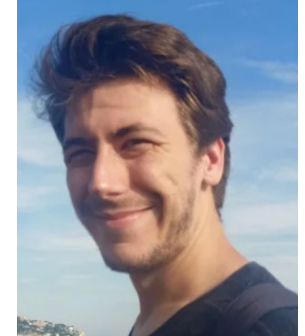


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Teaching Assistants & Research Engineer

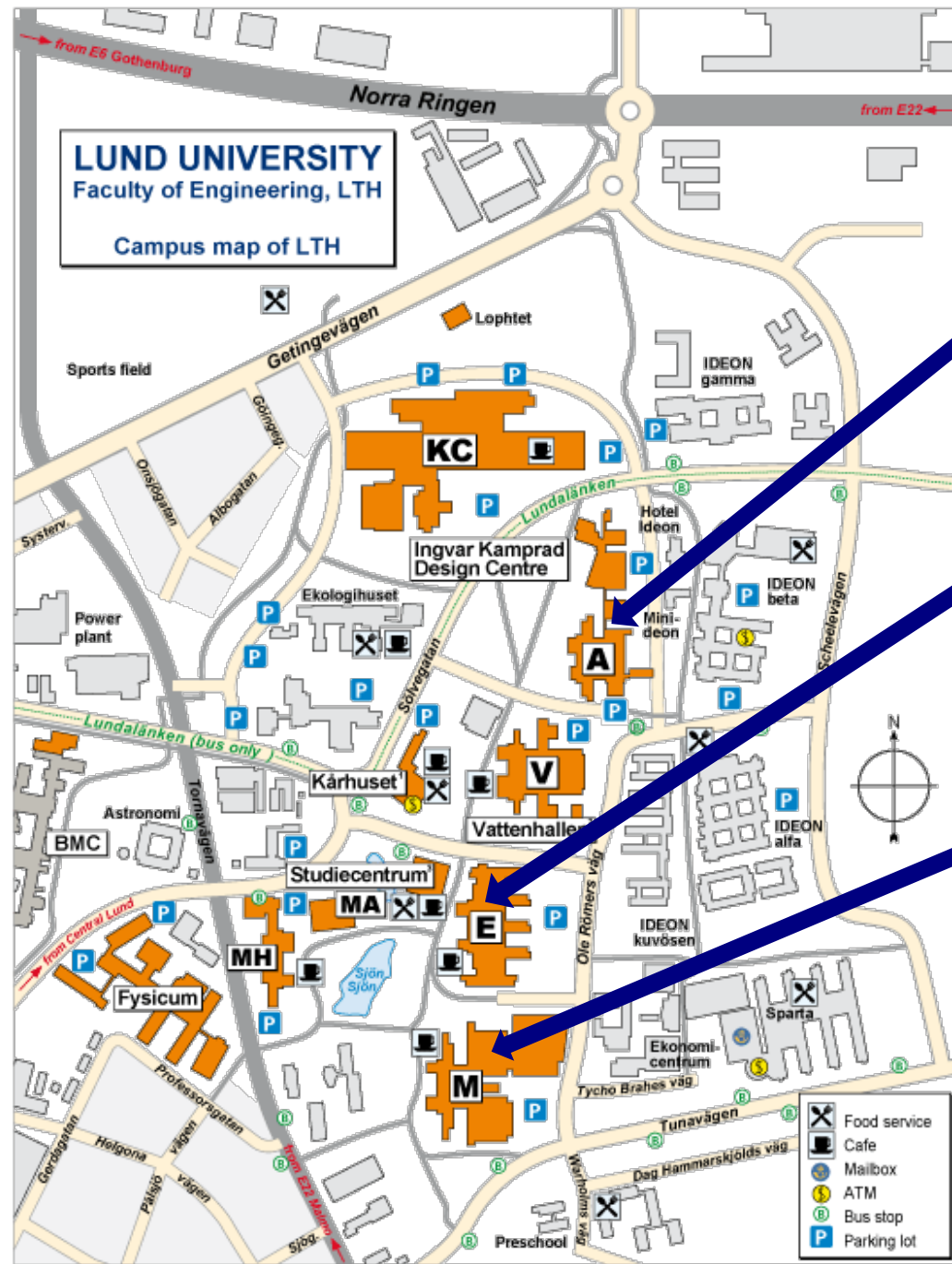
Exercises / Lab Exercises / Projects

- **Marko Guberina**
- **Sebastiano Fregnan**
- **Rasmus Kovács**
- **Anton Tetov Johansson**
- **Mathias Haage**
- **Alexander Pisarevskiy**
(research engineer)

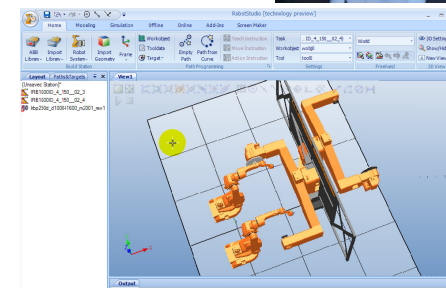


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see <https://www.control.lth.se/staff>

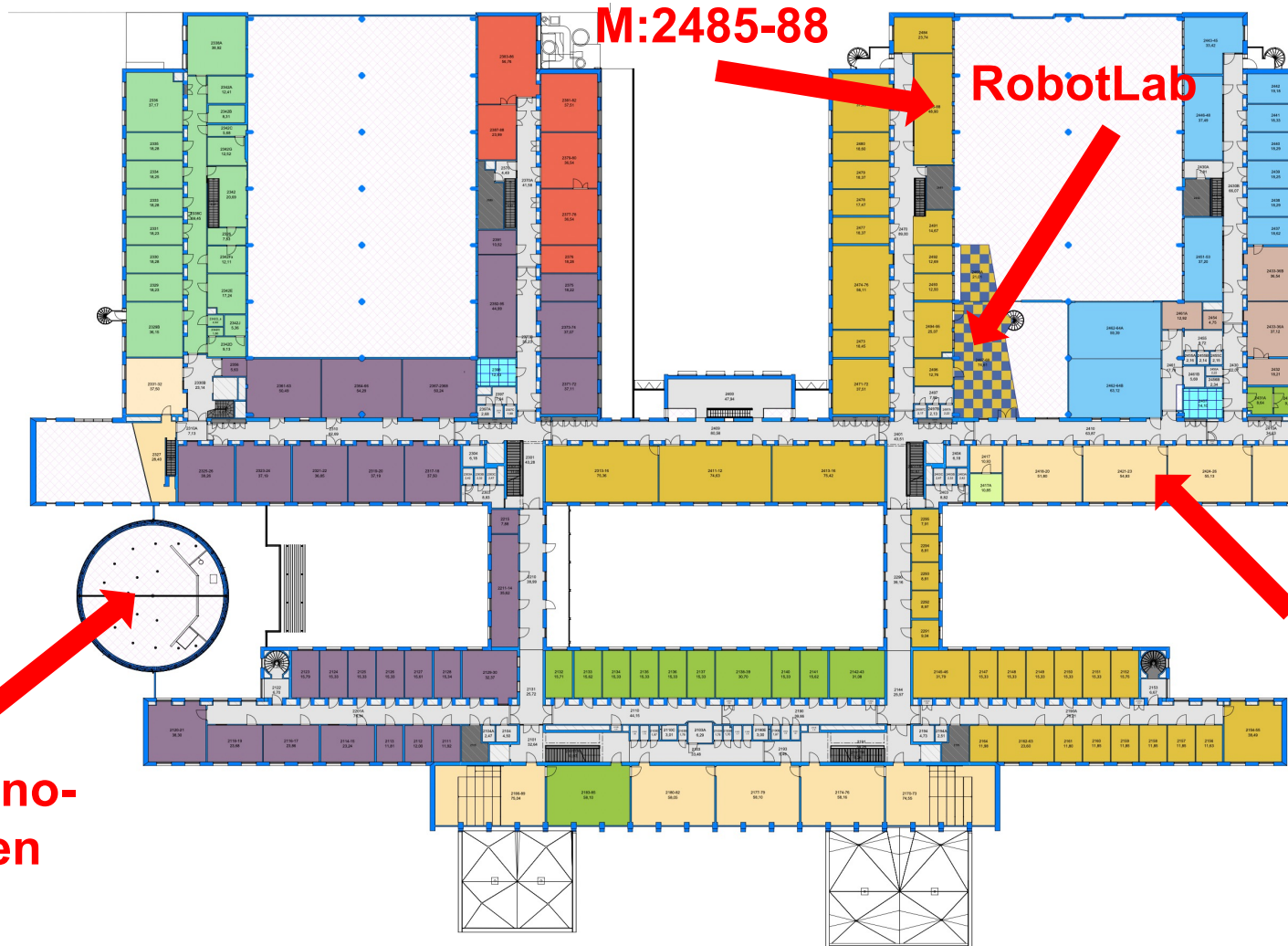


Student Union Building



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Mechanical Engineering Building, 2nd Floor



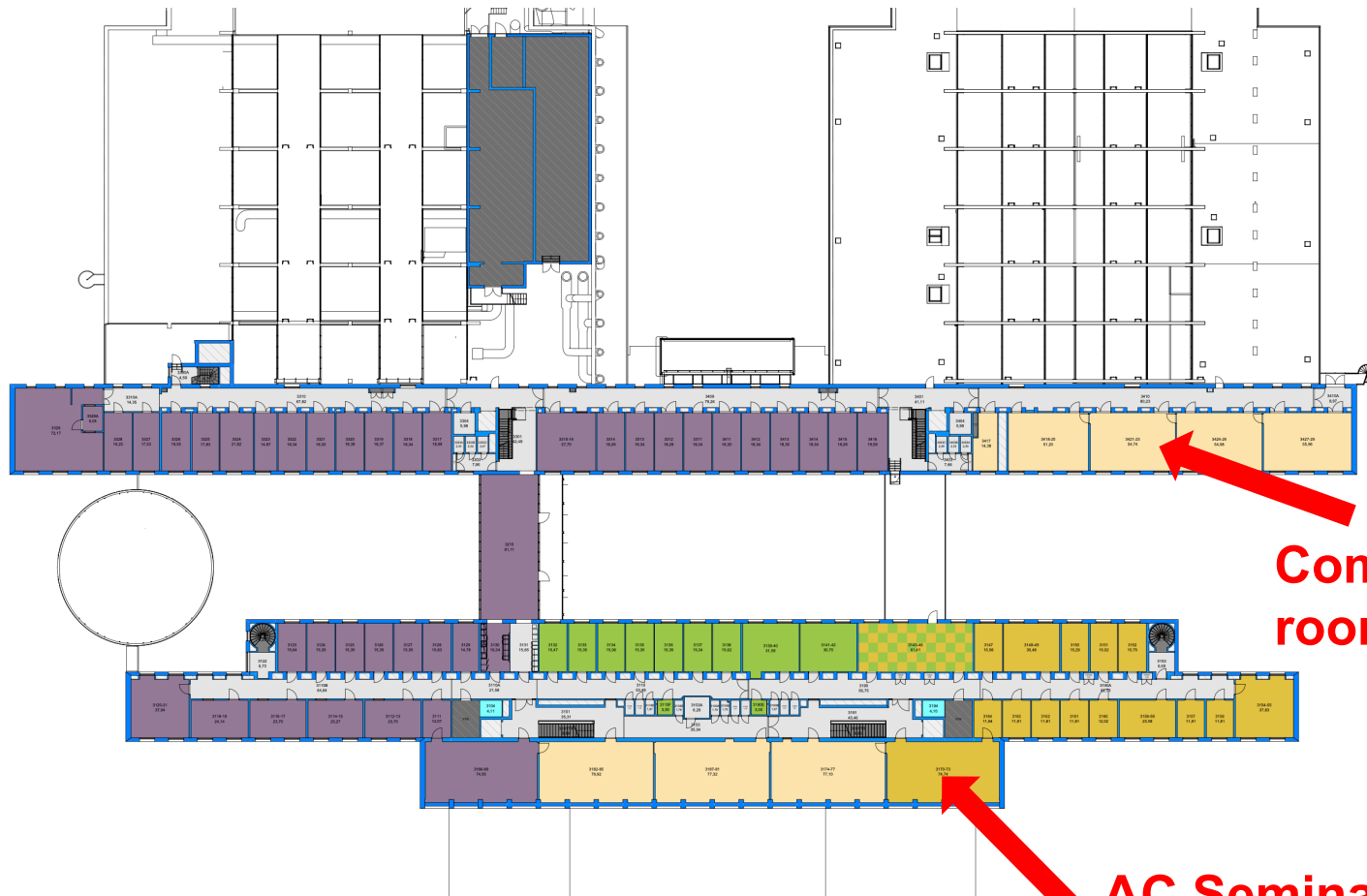
**M:Tekno-
dromen**

**Computer
rooms Ina**



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Mechanical Engineering Building, 3rd Floor



**Computer
rooms Emma**

**AC Seminar
Room M:O**



Course Program (see Canvas@LU)

Course Program: Applied Robotics (FRTN85) &
Applied Robotics for Architects (FRTN80),
7.5 credits, Fall 2026, Study Period 1

August 24, 2026

1 Course Staff

1.1 Course Coordinator and Examiner

- Björn Olofsson, Dept. Automatic Control, LTH
<https://www.control.lth.se/staff>

1.2 Course Administrator

- Mika Nishimura, Dept. Automatic Control, LTH
<https://www.control.lth.se/staff>

1.3 Teaching Assistants

- Sebastiano Fregnan, Dept. Automatic Control, LTH
- Rasmus Kovács, Dept. Automatic Control, LTH
- Anton Tetov Johansson, Dept. Architecture and Built Environment, LTH

1.4 Project Advisors

- Marko Guberina, Dept. Automatic Control, LTH
- Mathias Haage, Dept. Computer Science, LTH

1.5 Research Engineer

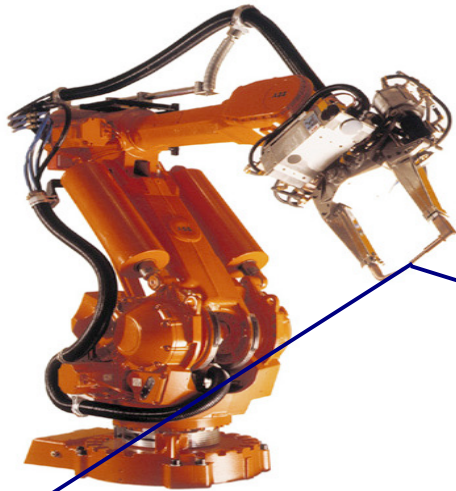
- Alexander Pisarevskiy, Dept. Automatic Control, LTH

2 Course Canvas

- <https://canvas.education.lu.se/>



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<https://canvas.education.lu.se>

The screenshot displays the Canvas LMS interface for the course 'FRTN85 & FRTN80'. The interface is framed by a blue line with bows at the corners, suggesting a 'house' shape. The left sidebar contains navigation links: Account, Dashboard, Courses, Calendar, Inbox, History, Studio, and Help. The main content area shows the course modules, including 'General Information' (Welcome, Schedule Fall 2026, Course Materials, Communication, Course Goals, Examination, Personnel) and 'Starting the Course' (Do this small "quiz" before you start, to unlock all material). The right sidebar includes links to View Course Stream, View Course Calendar, and View Course Notifications, along with a 'To Do' section showing a sign-up for a combined course.

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2026 HT/Autumn

Account

Dashboard

Courses

Calendar

Inbox

History

Studio

Help

Home

Assignments

Discussions

Grades

Syllabus

Quizzes

Modules

People

Canvas Survey

Notebook

FRTN85 & FRTN80 > Modules

Collapse All

General Information

Welcome

Schedule Fall 2026

Course Materials

Communication

Course Goals

Examination

Personnel

Starting the Course

Complete One Item

Do this small "quiz" before you start, to unlock all material

1 pts Score at least: 1.0

View Course Stream

View Course Calendar

View Course Notifications

To Do

Sign-up for combined course

FRTN85 Applied Robotics & FRTN80 Applied Robotics for Architects (2026)

Sep 3 at 1:15pm



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Theory and Practice

- Time and venues: see Canvas with link to TimeEdit
(need to sign-up by doing a “quiz”, log in with your LU-account)
- Lectures (at campus, recordings available from prev. years)
- Lab-exercises 1–3 (RobotStudio)
 - Hands-on exercise in RobotLab *this week*
- Computer exercises (kinematics, dynamics), **FRTN85**
- Robot system design feedback sessions, **FRTN80**
- **Optional** DC servo-lab, **FRTN85**
- **Optional** take-home exam for higher grades (grade 4, 5) **FRTN85**

Please register for the course in Ladok



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Examination – FRTN85

- Lab-exercises 1–3 (RobotStudio)
 - Hands-on exercise in RobotLab (week 1)
 - Hand-in exercise, kinematics & dynamics
 - Oral follow-up discussion on hand-in exercise
 - Project work; presentation and slides
 - **Optional** DC servo-lab
 - **Optional** take-home exam for higher grades (grade 4, 5) with oral follow-up discussion
- } pass / grade 3



Examination – FRTN80

- Lab-exercises 1–3 (RobotStudio)
 - Hands-on exercise in RobotLab (week 1)
- Hand-in exercise, robot system design for architectural task; short report
- Mandatory feedback seminars for hand-in exercise
- Project work; presentation and slides

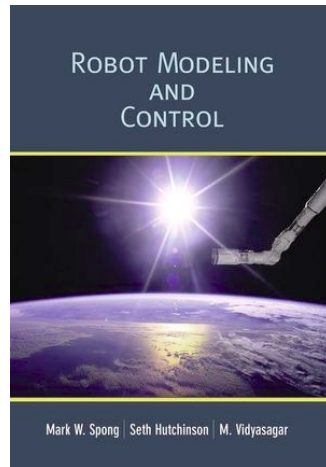
pass



Course Feedback

- Please provide feedback on the course, both during and after (CEQ)!
- Student representatives, **volunteers?**
- Updates since last year:
 - Revised kinematics / dynamics hand-in exercise in FRTN85
 - Updated lectures about dynamics in FRTN85
 - Revised feedback seminars for hand-in exercise in FRTN80
 - Oral follow-up discussion to hand-in exercise and take-home exam in FRTN85

Recommended Course Literature



**Spong, M.W., Hutchinson, S., and Vidyasagar, M.,
Robot Modeling and Control,
John Wiley and Sons, 2006**

**CONTROL METHODS
FOR ROBOTIC
APPLICATIONS**

Lecture notes

Leonid B. Freidovich

(June 9, 2017)



UMEA UNIVERSITY

Lecture notes by Leonid Freidovich

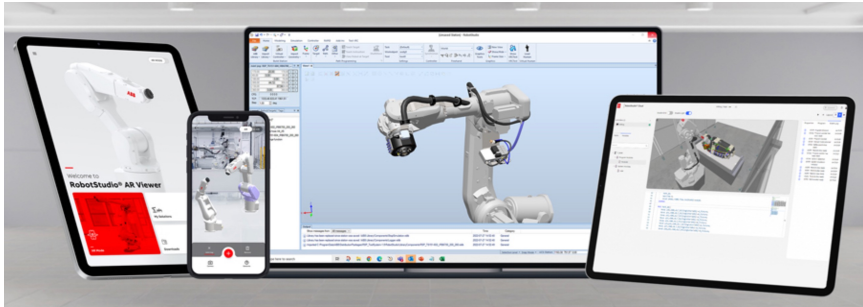
(based on Spong *et al.*)

Available on Canvas@LU



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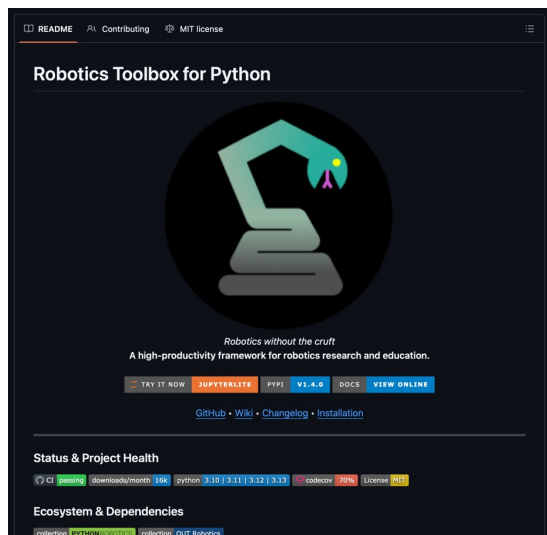
Software



RobotStudio

(PC-based robot simulation,
ABB Robotics)

<http://new.abb.com/products/robotics/robotstudio/>



Matlab & Python

Matlab version:

<https://petercorke.com/toolboxes/robotics-toolbox/>

Python version:

<https://github.com/petercorke/robotics-toolbox-python>

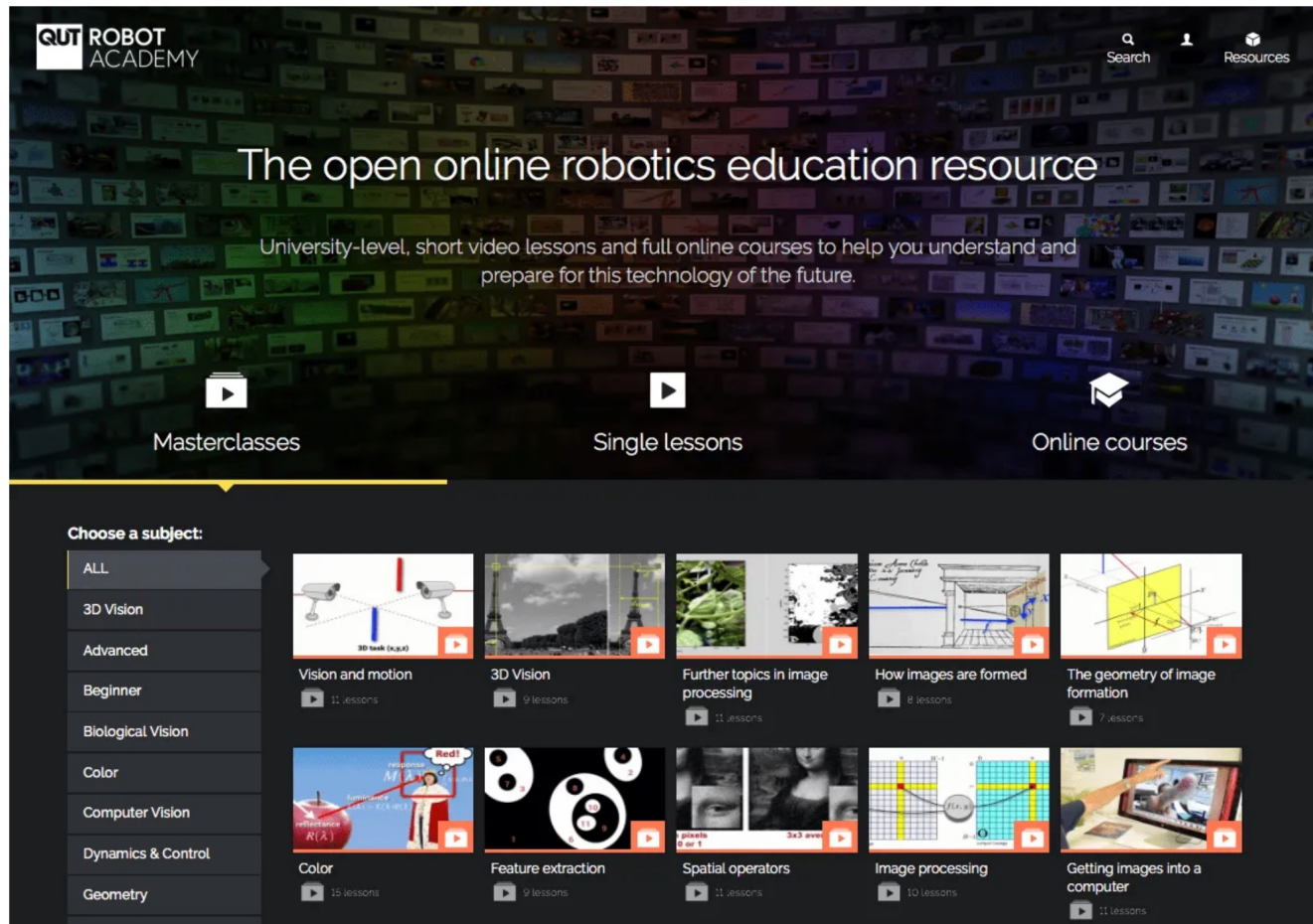
Accompanying books with theory and examples



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Optional Material (Recommended)

Peter Corke's Robot Academy

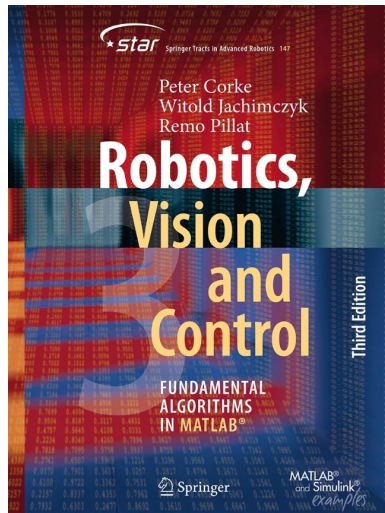


<https://petercorke.com/resources/robot-academy/>



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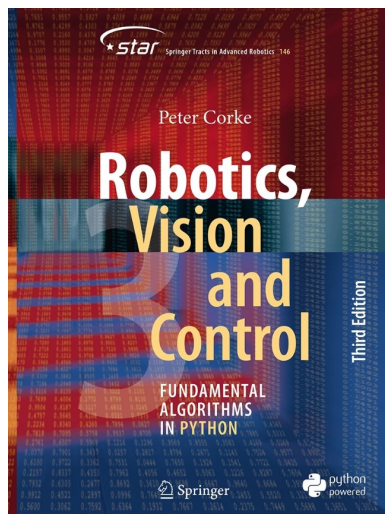
Optional Reading (Recommended)



Books for Peter Corke's robot toolbox

Available from LU network (and with VPN)
as e-book at

<https://doi.org/10.1007/978-3-031-07262-8>



Python version also available (though not
accessible from LU network):

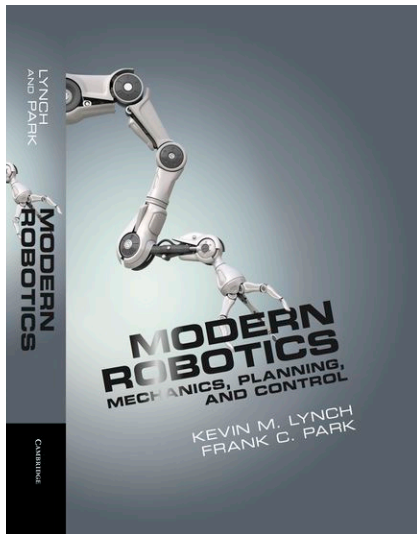
<https://doi.org/10.1007/978-3-031-06469-2>



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Optional Reading (Recommended)

“Modern Robotics: Mechanics, Planning, and Control”
by Kevin M. Lynch and Frank C. Park



- Available as online pdf book
- Video lectures
- Rotations based on screw-theory and exponential representations
- More mathematical presentation

http://hades.mech.northwestern.edu/index.php/Modern_Robotics



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Compulsory Hands-On Exercise

Please sign-up for the first combined computer exercise and **compulsory** hands-on exercise in the RobotLab.

Alternatives:

Thursday Sep. 3, 13:15–15,

Thursday Sep. 3, 15:15–17,

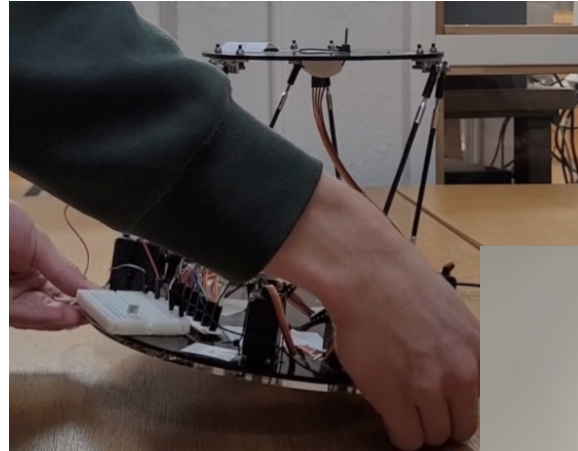
Friday Sep. 4, 8:15–10,

Friday Sep. 4, 10:15–12

Assignment in Canvas with instructions for **sign-up**

Preparation: Read hand-out before coming to the lab!

Projects – Examples from Previous Years



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Project Seminar

- The course project seminar is scheduled for
 - Friday October 23, 8-12
- Please let me know as soon as possible if you have other mandatory exams colliding with this time.



Lund Formula Student

Announcement from **Lund Formula Student (LFS)**:

- They are recruiting for the 2027 team
 - Design and build a driver-less, electric racing car
- Apply knowledge from this and other courses in practice
- More information at <http://lundformulastudent.se/>



Introduction to Robotics

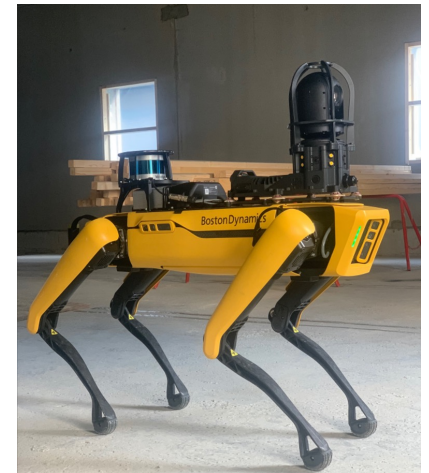
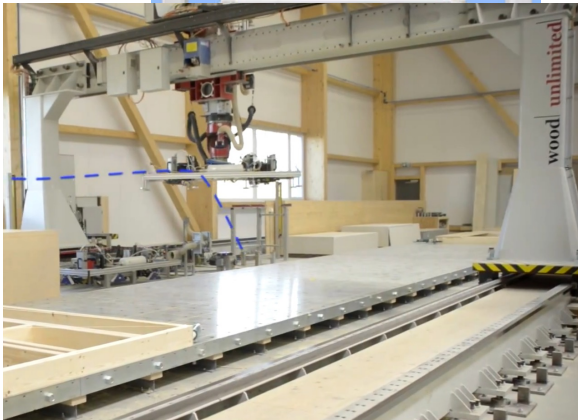


Robots – What Kinds of Robots?



Industrial robotics
Mobile robotics
Service robotics
Entertainment

...



Multi-disciplinary:

**Nonlinear control, mechatronics, real-time embedded systems,
software, machine learning, design, sensors, ...**



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Some Robot Classifications

- Service Robots
 - Robot vacuum cleaners
 - Lawn mowers
 - Restaurant service robots
- Industrial Robots
 - Serial-type robots
 - Parallel kinematic machines
 - Example applications: welding, spray painting, grinding, polishing, and milling
- Legged Robots (Service, Industrial, and “Entertainment”)
 - Quadrupedal robots, “robot dogs”
 - Humanoids

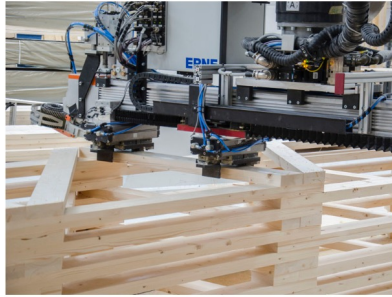




Real-time coordination in collaborative machining

Lund University and Güdel AG exhibit real-time coordination between robots with significantly different types of kinematics and control systems. Requiring different robots to work together is an example of the heterogeneous situation that is typical at SMEs. The state of the art motion-coordination software is demonstrated by collaborative machining of parts for wooden boxes. Here, a Güdel parallel-kinematic concept robot and a standard ABB serial-kinematic robot complement each other well to solve the task: The parallel robot offers exceptional stiffness and accuracy for machining, and the serial robot can perform both handling and rough-cut machining. These two relate to the project's demonstrator currently deployed at a Swiss woodworking company, with software services being loosely coupled for flexible configuration while supporting tight real-time control loops for efficiency during production.





Robot-Based Digital Fabrication, Gramazio Kohler Research, ETH Zürich

The sequential roof

<https://gramaziokohler.arch.ethz.ch/web/e/projekte/201.html>

Semiramis

<https://gramaziokohler.arch.ethz.ch/web/e/projekte/409.html>

Printed structures

<https://gramaziokohler.arch.ethz.ch/web/e/forschung/451.html>



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Medical Robotics

https://www.youtube.com/watch?v=7sTfD_mStwE

da Vinci-robot by Intuitive surgery

["Stability of Haptic Obstacle Avoidance and Force Interaction",
R Johansson, M Annerstedt, A Robertsson (2009)]



da Vinci Surgeon
Less invasive • More precise • Faster recovery

Cardiothoracic General Surgery Gynecology Head & Neck

Home > Surgeon Locator > Surgeon Profile

EMAIL THIS PAGE PRINT

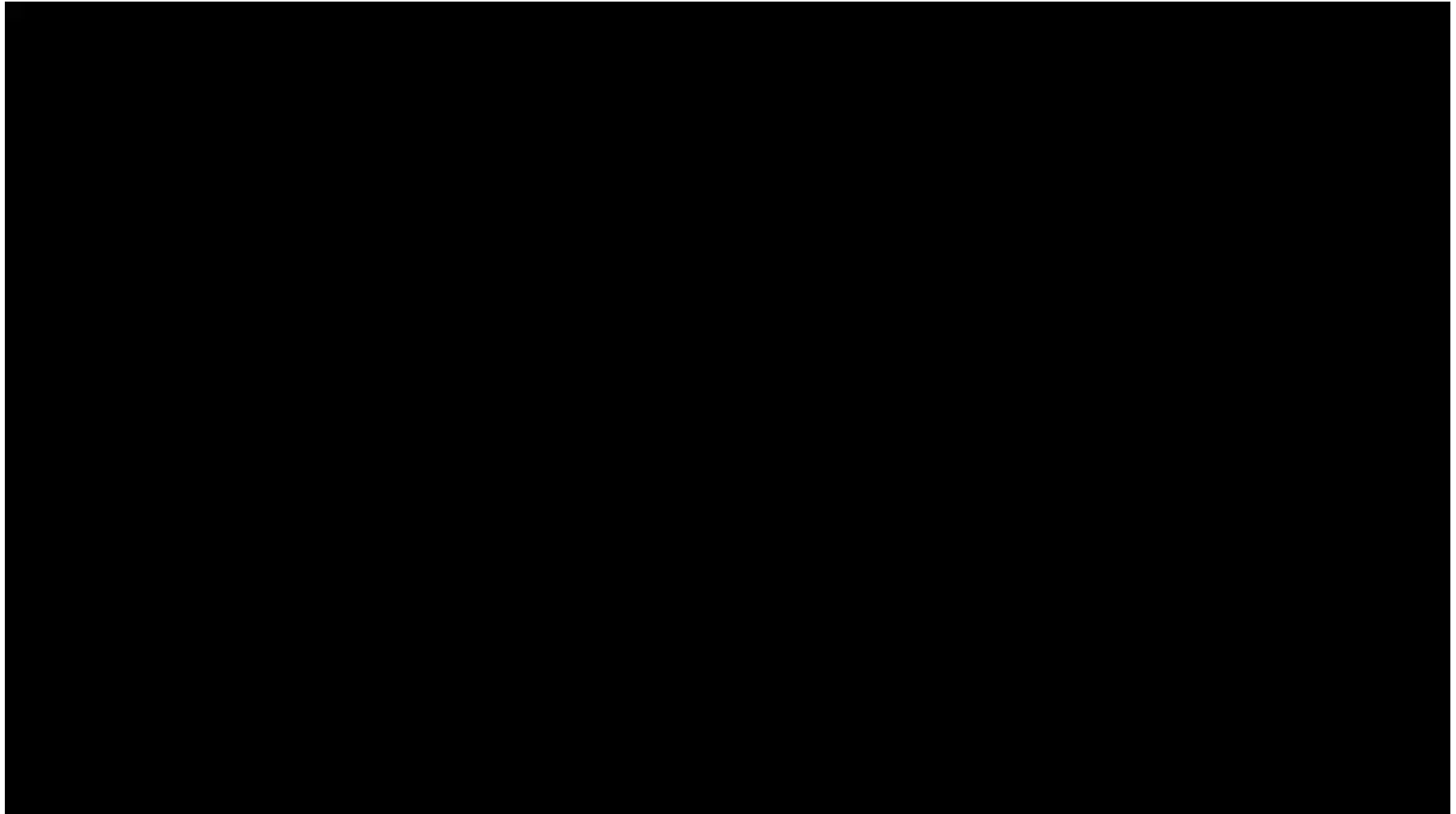
Surgeon Profile

Name	Dr. Magnus Annerstedt, Urologist
Address	Herlev Ringvej 2730 Herlev, , Denmark
Website	
Specialty	Urology Prostatectomy Kidney Cancer, Bladder Cancer
Contact	✉ Contact This Surgeon

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<https://www.youtube.com/watch?v=C17-bGquljl>

Dual-Arm Haptic Robot Control



Robots and Application Areas



Robotics in This Course

- The following conceptual problems must be resolved to make a robot succeed in performing a typical task:
 - Forward Kinematics
 - Inverse Kinematics
 - Velocity Kinematics / Jacobians
 - Dynamics
 - Path Planning and Trajectory Generation
 - Motion Control
 - (Force Control)
 - Sequence programming (and task description)



Robotics – Some Characteristics

- The application, tooling, sensors, design of robots, ...
- **Robot system design** for solving a certain task
- Requires consideration of several factors for **realizing a functioning system**



Degrees of Freedom

- An object has n degrees of freedom (DOF) if its configuration can be minimally specified by n parameters.
- The number of DOF is equal to the dimension of the configuration space.
- For a robot manipulator:

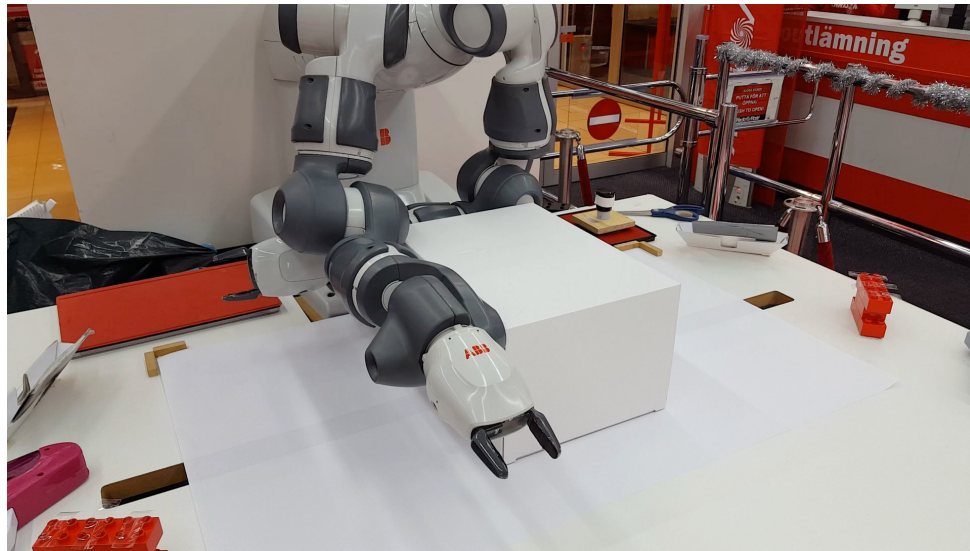
number of joints = number of DOF

ABB YuMi



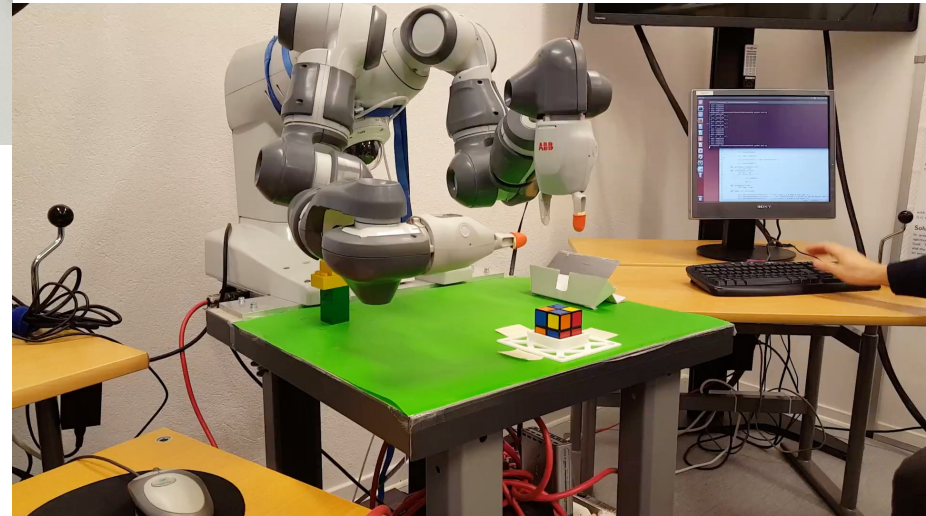
Degrees of Freedom – Examples

The GiftWrapper & Rubik's Cube

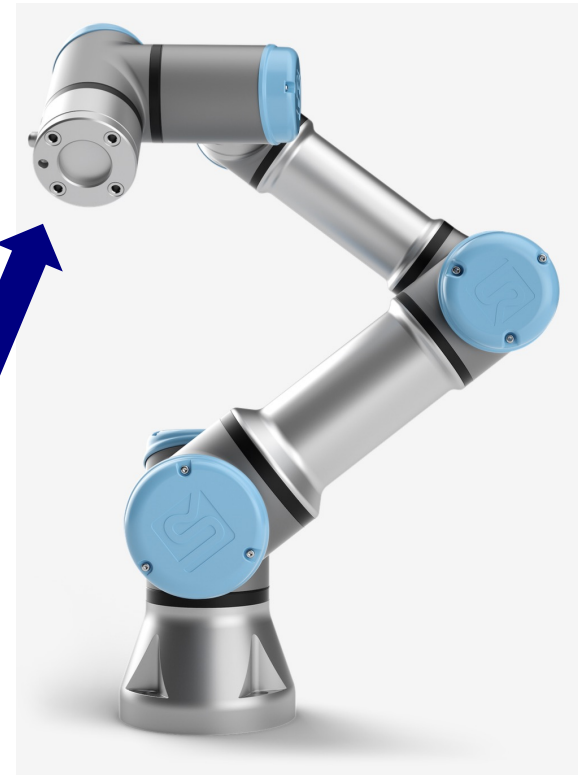
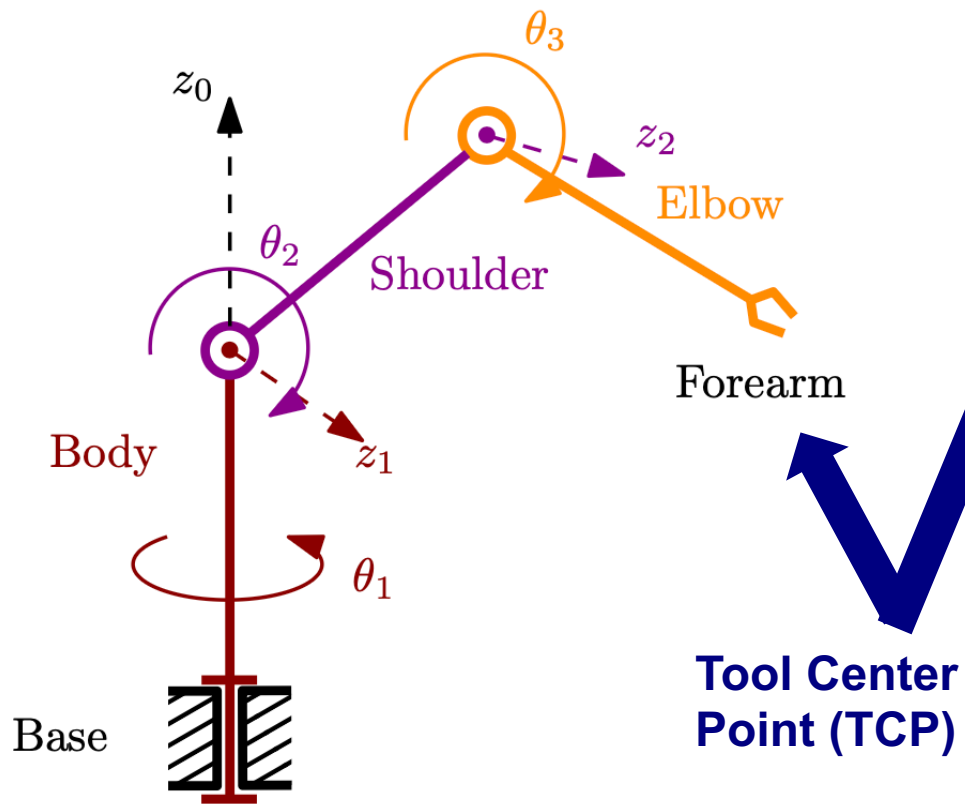


Stenmark, M., Stolt, A., Topp, E. A., Haage, M., Robertsson, A., Nilsson, K., & Johansson, R. (2016). "The GiftWrapper: Programming a Dual-Arm Robot With Lead-through"

Course project for solving the Rubik's cube using vision and a dual-armed robot



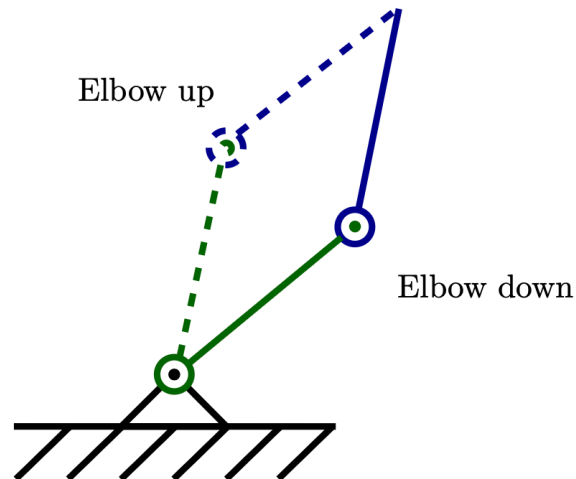
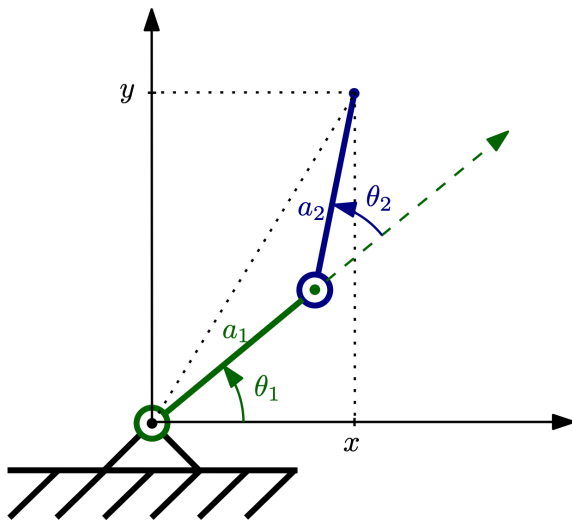
Robot Arm



Forward and Inverse Kinematics

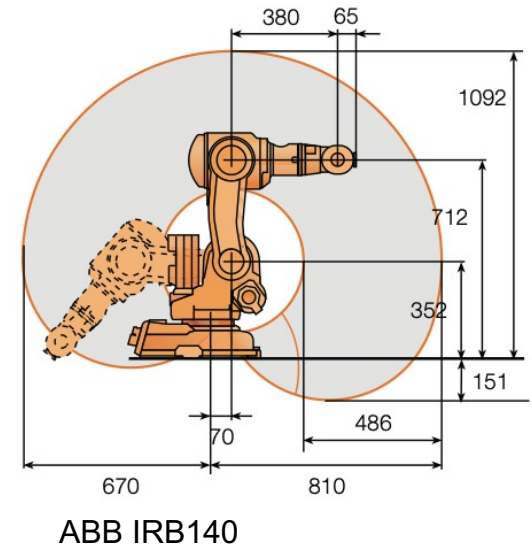
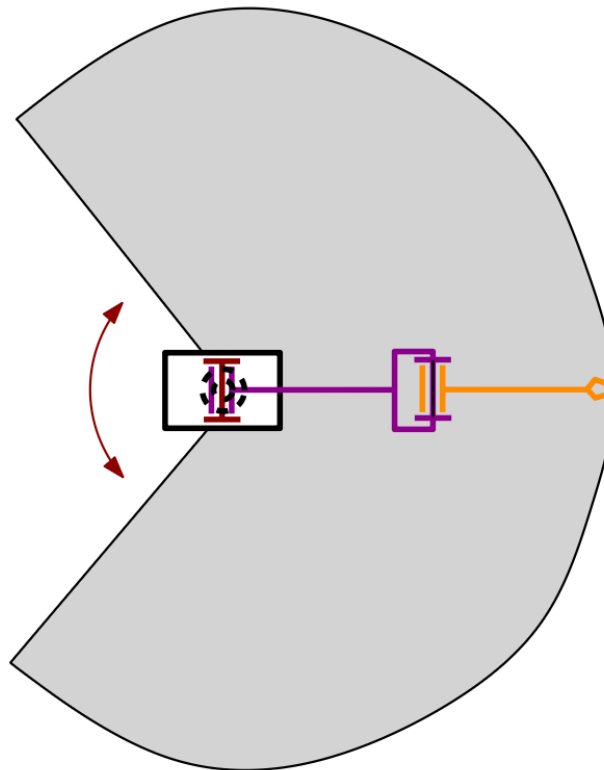
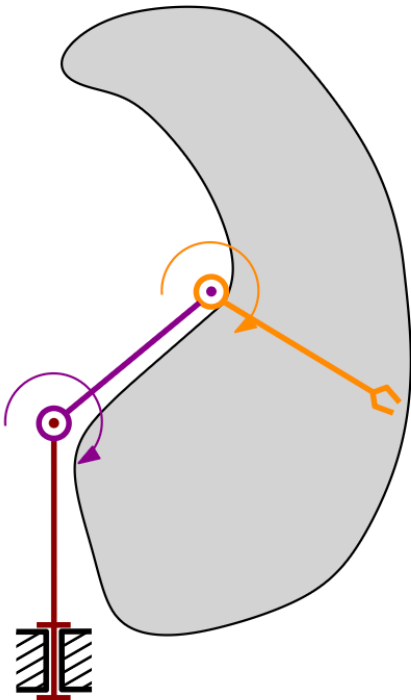
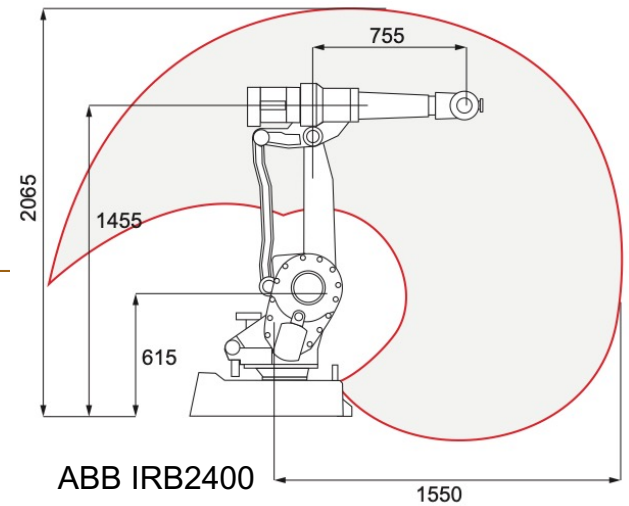
- **Forward kinematics:**
 - Given joint angles, find tooltip pose (*pose: position and orientation*)
- **Inverse kinematics:**
 - Given desired tooltip pose, find joint angles

Possibly several different solutions



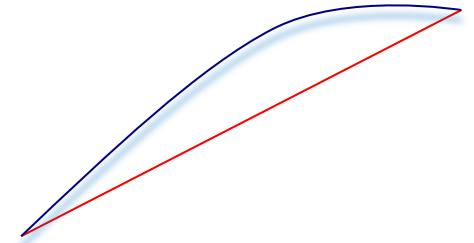
Workspace

- Joint limitations taken into account



Robot Motions

- Point-to-point motion
 - **MoveL** - Moves the tooltip (TCP) of the robot linearly
 - **MoveJ** – “joint interpolation” (usually ends up with *curved Cartesian motion*)
- Path generation
 - Geometric path (curve in space)
- Trajectory tracking
 - Geometric path AND time matters
 - At what time are you in what position with what velocity/acceleration, etc.



Ethical Questions to Consider

- Should a company not use robots to replace workers, unless they are forced to by global competition? What is your opinion? **[Discuss 2 min]**
- A robot could do hazardous and strenuous jobs that workers cannot or do not want to do.
- Robotics can create new jobs – in engineering, science, design, architecture – and save jobs in high-cost countries.
- Robotics can increase product quality and repeatability.
- Robotics is finding new applications in the domestic service market.



Deburring in ProFlexA Project

**ProFlexa Project
Productive & Flexible Automation**

**Grinding Demonstrator at
Combi Wear Parts, Ljungby Sweden**

IFR International Federation of *Robotics*



Menu

- Home
- Association
- History
- Industrial Robots
- Robots Create Jobs
 - Work Unsafe vor Humans
 - Work in High Wage Countries
 - Work Impossible for Humans
- Service Robots
- Robotics Research
- Standardisation
- News
- CEO Statements
- Events
- Downloads

Robots Create Jobs

ROBOTICS will be a major driver for global job creation over the next five years. The announcement is based on a study conducted by the market research firm, Metra Martech, "Positive Impact of Industrial Robots on Employment".

One million industrial robots currently in operation have been *directly* responsible for the creation of close to three million jobs, the study concluded. A growth in robot use over the next five years will result in the creation of one million high quality jobs around the world. Robots will help to create



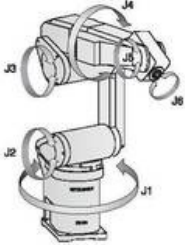
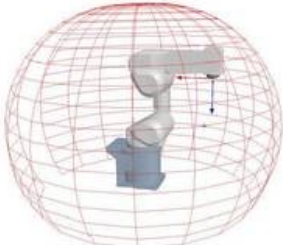
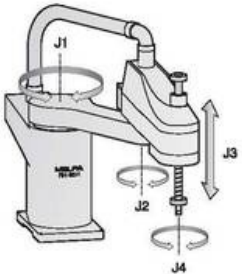
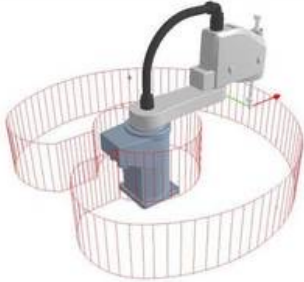
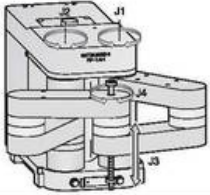
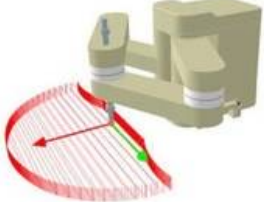
Industrial Robot

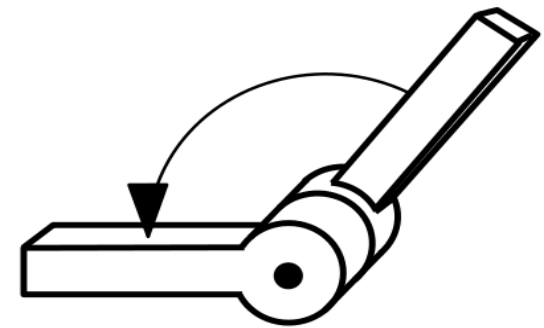
Industrial robot as defined by ISO 8373:

An **automatically controlled, reprogrammable, multipurpose manipulator programmable in three or more axes**, which may be *either fixed in place or mobile* for use in industrial automation applications.



Robot Designs (1/2)

Principle	Kinematic Structure	Photo
Articulated Robot 		
SCARA Robot 		
SCARA Robot 		

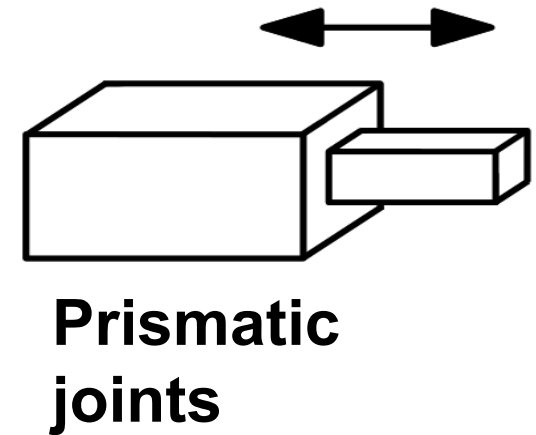
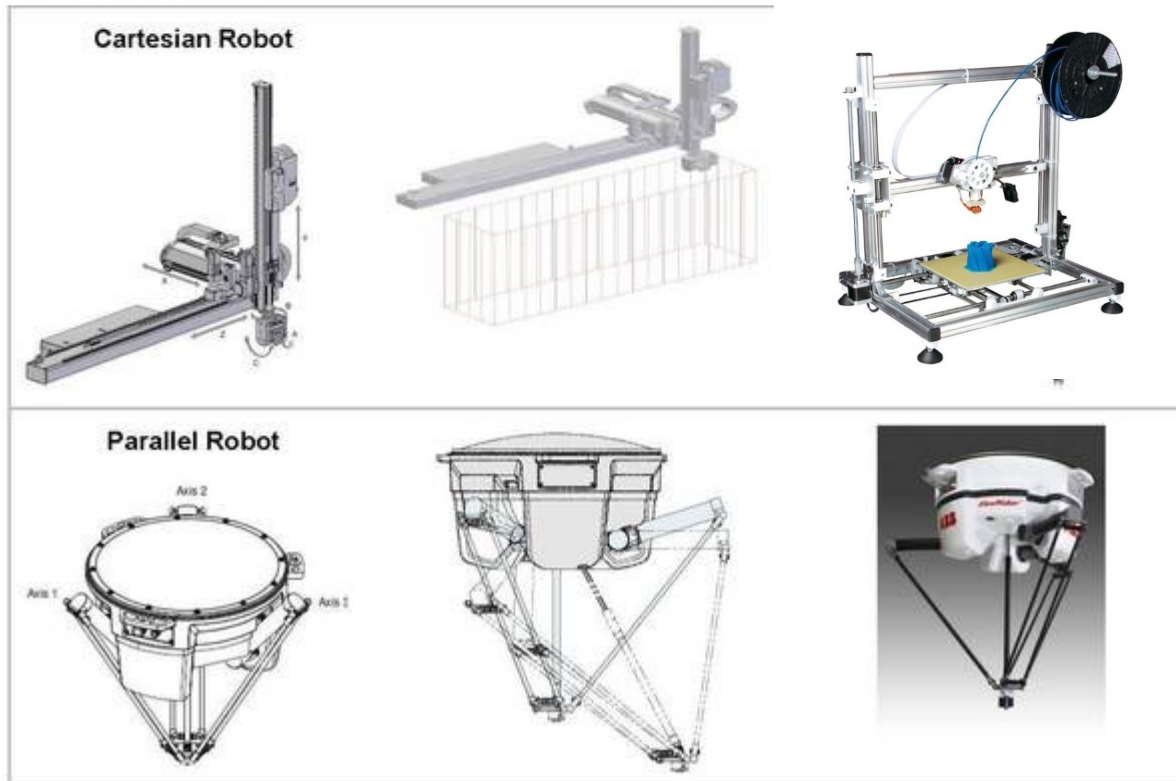


Revolute joints



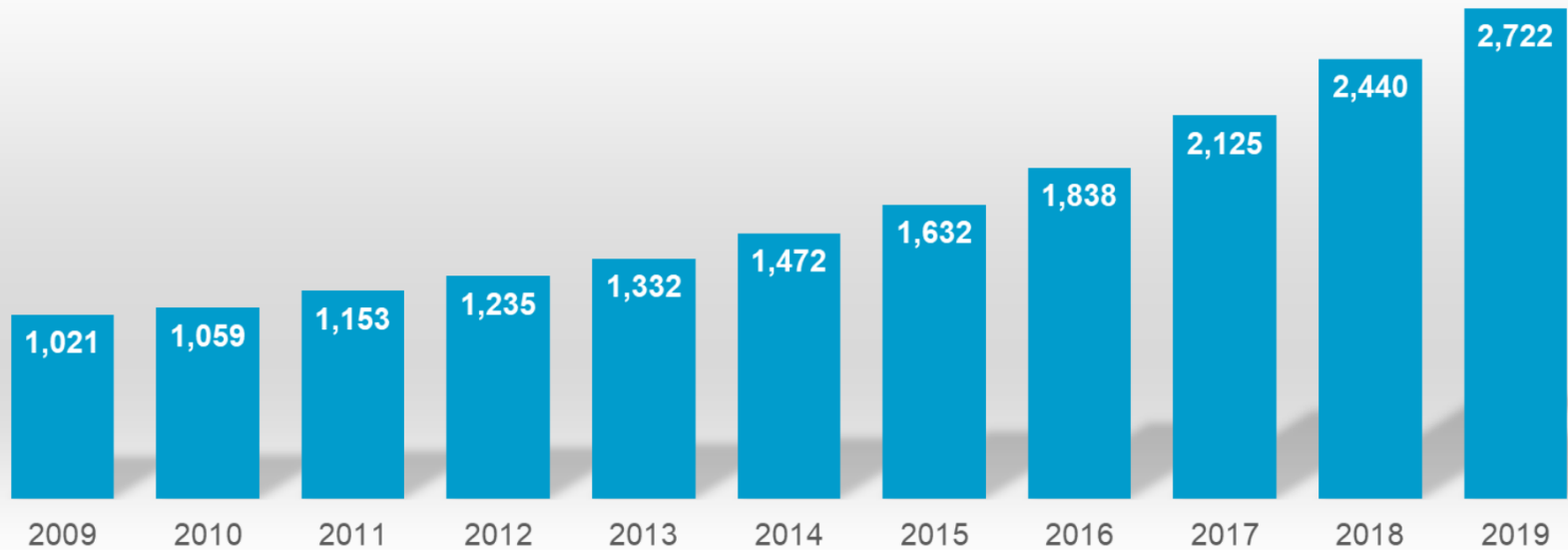
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Robot Designs (2/2)



Some Statistics (1/2)

Operational stock of industrial robots - World
1,000 units

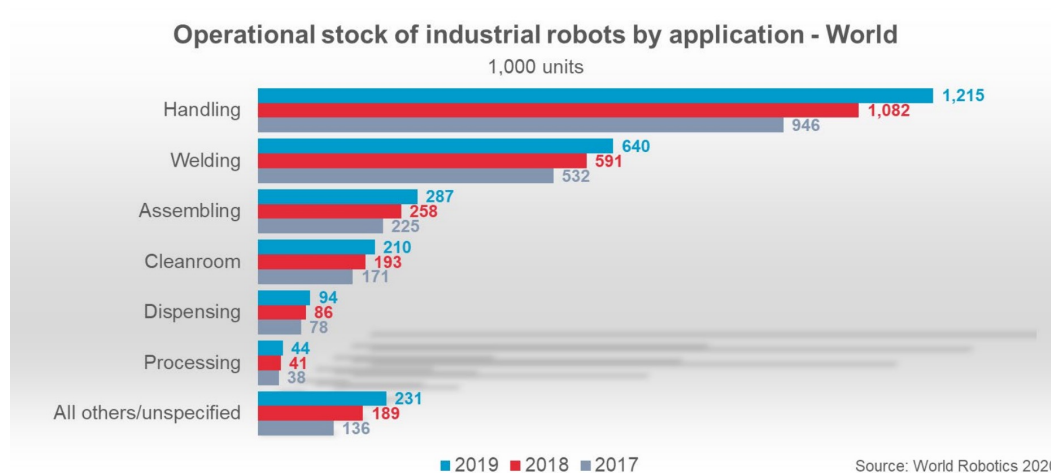
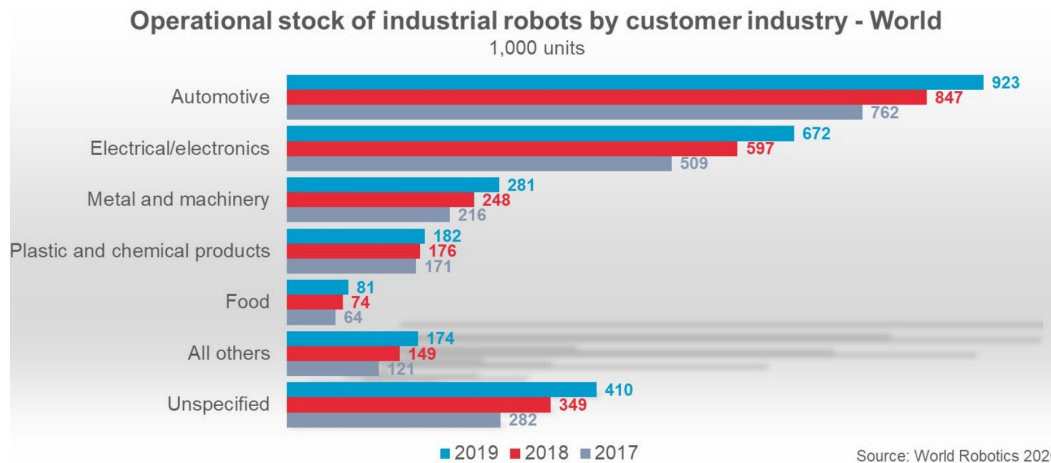


Source: World Robotics 2020



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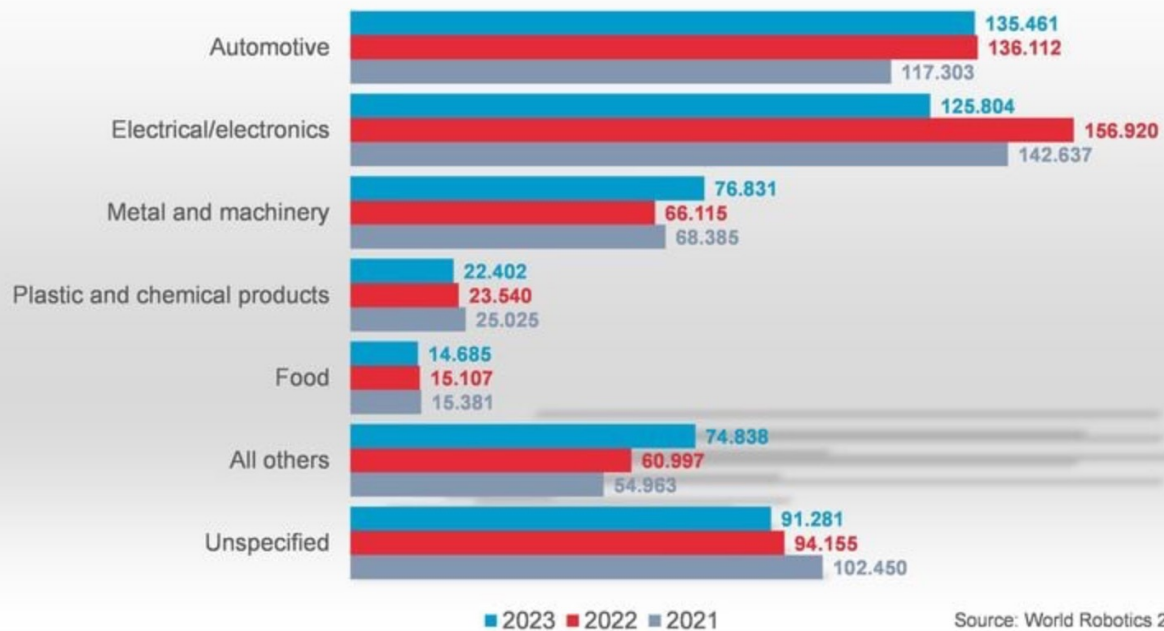
Some Statistics (2/2)



Industrial Areas



Annual installations of industrial robots by customer industry - World
units



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Automotive Example: Production of Cars



<https://www.youtube.com/watch?v=u2BfVmmaWvk>



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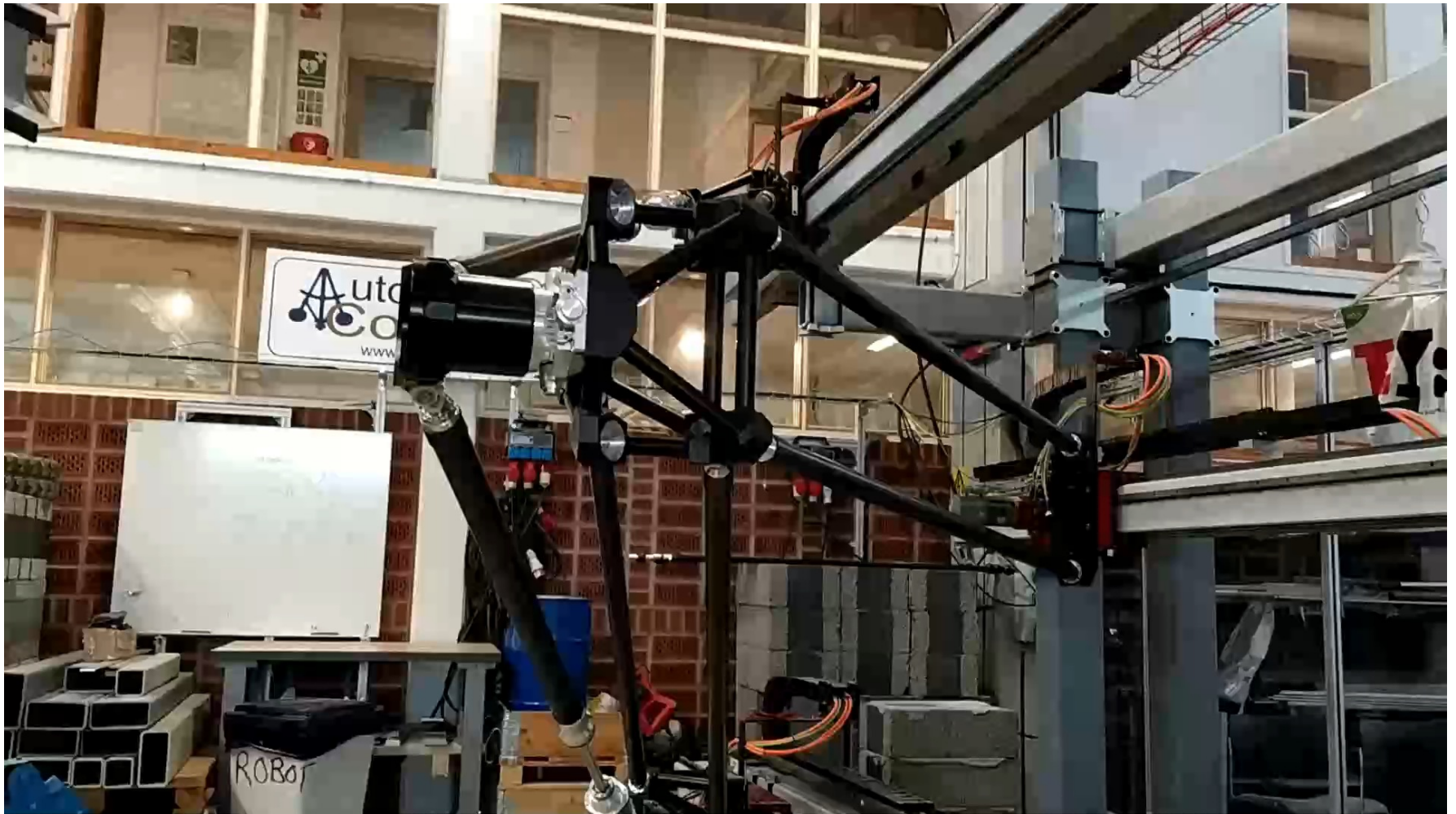
New Industrial Robot Designs



- Control engineers and mechanical design engineers must work together at concept stage (controller tuning stage too late)
- Parallel (stiff) designs increase resonance frequencies
- Collaborative robots – design and control
- Control structures should be tested on elasticity models already at concept stage (e.g., Modelica)

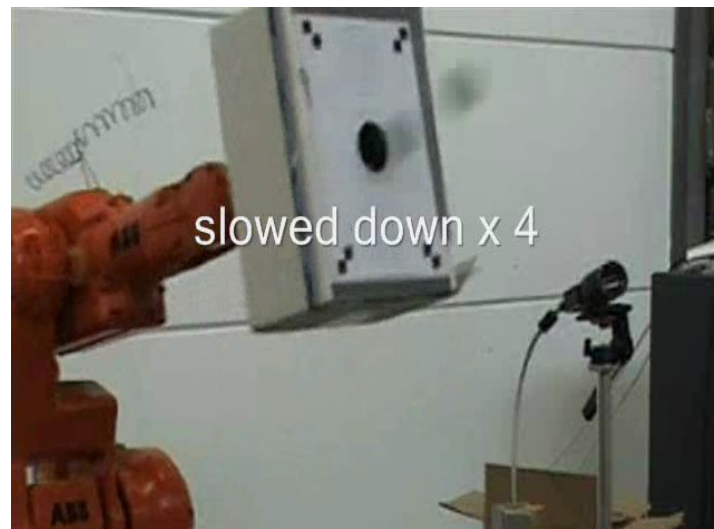


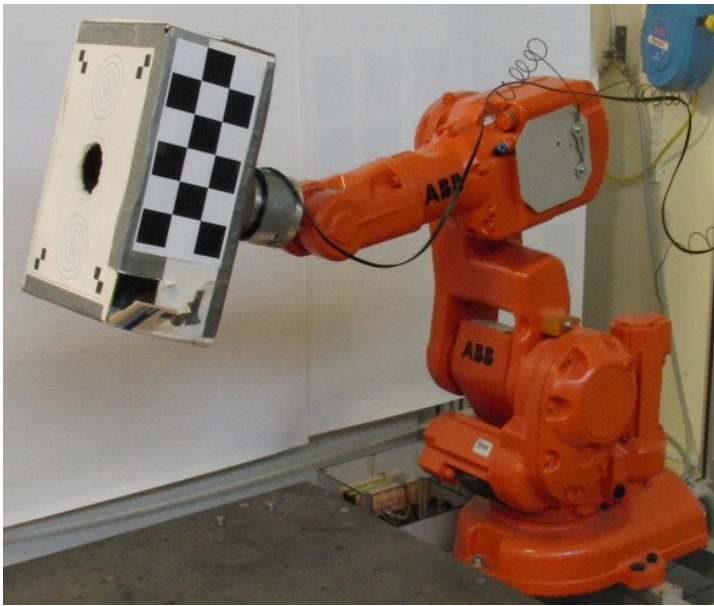
New Industrial Robot Designs (cont'd)



Force and Vision – Reaction Time Crucial

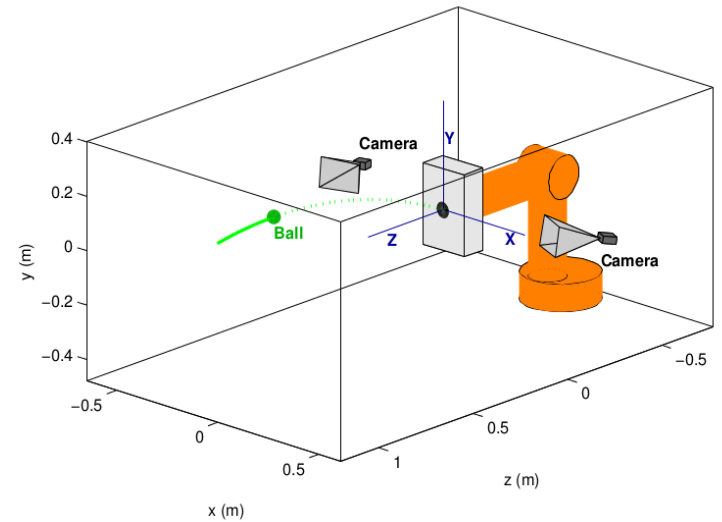
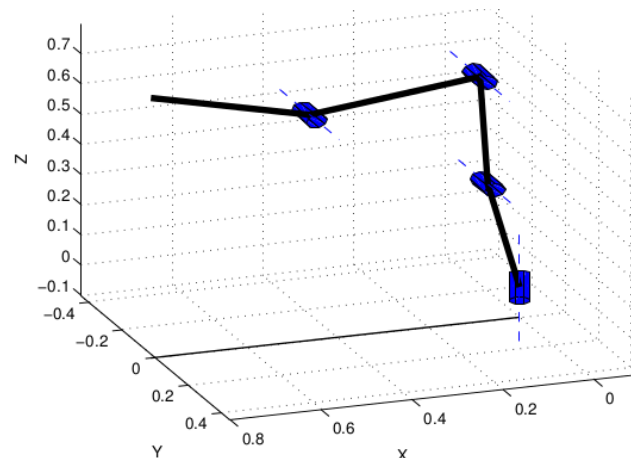
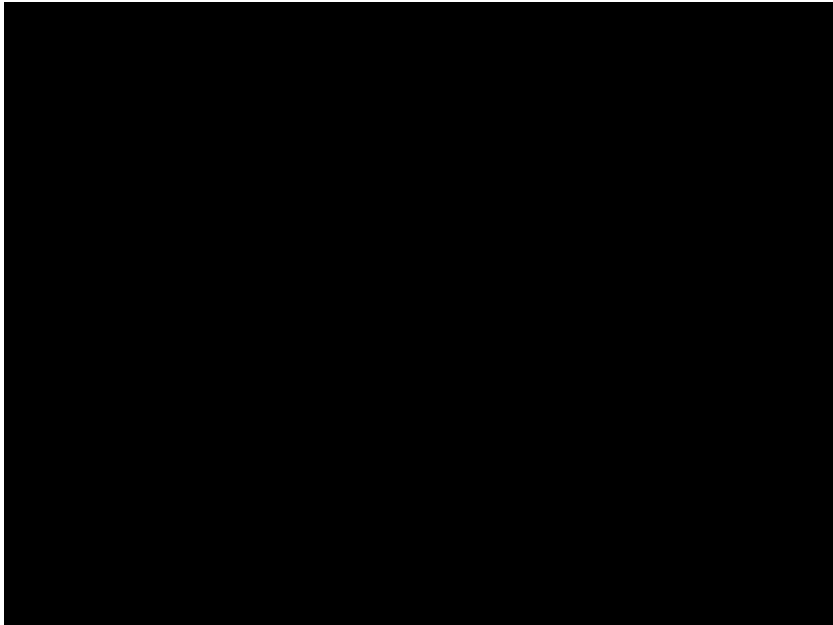
Applications and
enabling technology ...





Ball-and-dart-catching robot

<https://www.youtube.com/watch?v=XP7yWhN6V-k>



Linderoth, M., Robertsson, A., & Johansson, R. (2009). "Vision based tracker for dart catching robot"

Ardakani, M. M. G., Olofsson, B., Robertsson, A., & Johansson, R. (2018). "Model predictive control for real-time point-to-point trajectory generation"



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Summary



Summary

- Course administration
- Challenges of robotics (interdisciplinary, design, systems perspective)
- Different applications of robotics – wide range of areas
- Recommended reading:
 - Lecture notes Chapter 1





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